# Predictors of COVID-19 Protective Behavior Before and After Mask Mandates: A Cross-National Analysis

Zixuan Zhao A1948883

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# Background & Motivation

## 1. Project background

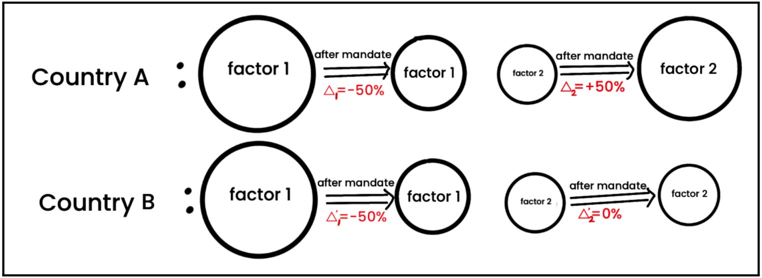
- During **COVID-19**, protective behaviours such as **mask wearing**, social distancing, hand hygiene, and self-isolation were important for reducing transmission.

- Compliance was shaped not only by policy rules, but also by risk perception, trust, demographic characteristics, wellbeing, and broader social context.

- These predictors may operate **differently across countries** because policy environments, institutions, culture, and public expectations vary.

## 2. Research questions

- Within each country, do the key predictors of compliance with COVID-19 protective behaviours change before and after mask mandates?
- Are these changes consistent across countries, or do they vary?
- Is there a set of common cross-national predictors, alongside country-specific predictors?



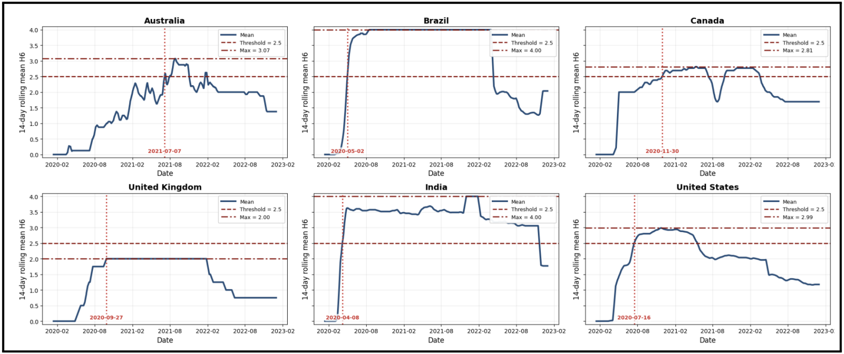
# Data Sources

## 1. Behaviour and policy datasets

- **YouGov dataset:** repeated survey responses about people's COVID-19 behaviours, attitudes, and background characteristics.
  - Behaviour: whether people wore masks, kept physical distance, washed hands...
  - Predictor: age group, gender, confidence in health services...
- **OxCGRT dataset:** This dataset contains the strictness with which the government has enforced its mask-wearing policy.
- The two sources are linked by **country, region and survey date**.

# Data Sources

## 2. Policy timing figure




This figure show how mask policy intensity changed over time, and how the mandate threshold was used to define the before and after mandate periods.

# Data Sources

## 3. Country coverage

The current cleaned cross-national sample includes six countries:

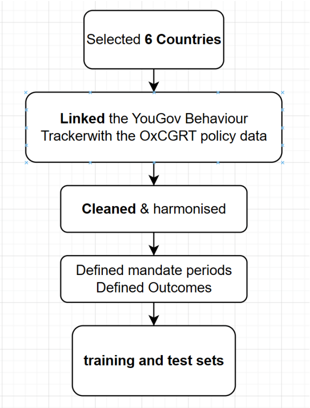
| Country code | Country name |
|---|---|
| AUS | Australia |
| BRA | Brazil |
| CAN | Canada |
| ~~CHN~~ | ~~China~~ |
| GBR | United Kingdom |
| IND | India |
| USA | United States |

China was excluded from the current analysis because several key variables needed for the harmonised workflow were not available.

# What Have I Done So Far?

## 1. Data pre-processing

- For the YouGov and OxCGRT files, various formatting issues have been resolved.
- **Mapped survey regions to policy regions**: Convert all place names in both datasets to lower-case English letters, then match them one-to-one.
- **Defined before/after mandate periods**.
- **Defined two outcomes**: Face-mask-wearing and Overall-protective-behaviour.
- **Cleaned the data**: Removing columns and samples with too many missing values, encoding the data, making column names more readable, and splits.



## 2. About Model

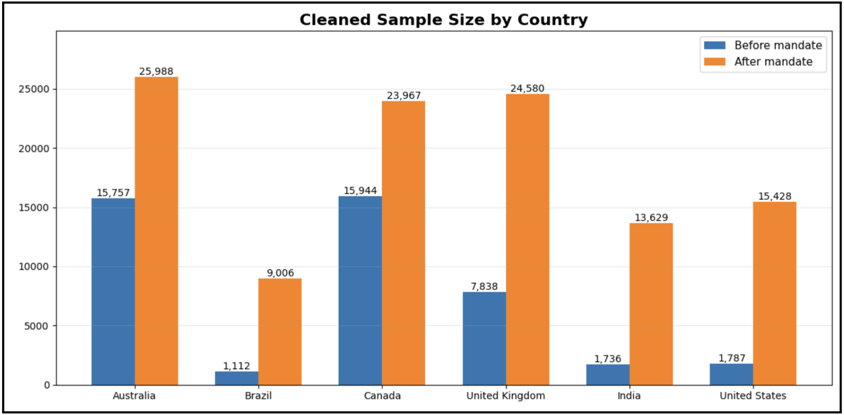
**Model 1: Random Forest**
This model builds many decision trees and lets them vote for the final prediction. Capable of handling non-linearity.

**Model 2: XGBoost**
This model also uses decision trees, but it builds them step by step. Each new tree tries to correct the mistakes made by the previous trees. Capable of handling non-linearity.

**Metric: AUC**
AUC means Area Under the ROC Curve; selected because it can measure model separation across all classification thresholds. AUC ranges from 0 to 1. A higher AUC indicates better separation between compliant and non-compliant respondents.

## 1. Cleaned sample size

The preprocessing pipeline produced cleaned before- and after-mandate datasets for all six countries.



## 2. Train-test split

The next step was to prepare data for modelling.

- I split the data separately for each country, policy period, and outcome.
- I used an 80/20 train-test split.
- Stratification (based on the outcome class, maintain a similar proportion of categories in the train and test sets) was used where possible to preserve the class balance of the outcome.

| Dimension | Values | Number |
| --- | --- | --- |
| Countries | AUS, BRA, CAN, GBR, IND, USA | 6 |
| Policy periods | Before mandate, after mandate | 2 |
| Outcomes | Face mask, overall protective behaviour | 2 |
| Total modelling tasks | $6 \times 2 \times 2$ | 24 |

# Results

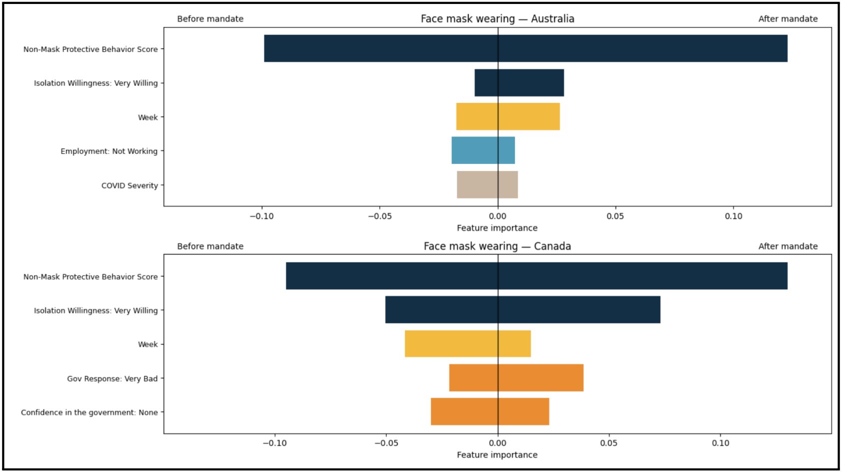
## 3. Model Performance (AUC)

| Country | Outcome | RF | XGboost |
|---|---|---|---|
| Australia | Face mask wearing Before mandate | 0.844 | **0.854** |
| | Face mask wearing After mandate | 0.909 | **0.912** |
| | Protective behaviour Before mandate | 0.767 | **0.772** |
| | Protective behaviour After mandate | 0.826 | **0.831** |
| Brazil | Face mask wearing Before mandate | 0.783 | **0.785** |
| | Face mask wearing After mandate | 0.652 | **0.653** |
| | Protective behaviour Before mandate | **0.713** | 0.705 |
| | Protective behaviour After mandate | **0.829** | 0.827 |
| Canada | Face mask wearing Before mandate | **0.854** | **0.854** |
| | Face mask wearing After mandate | 0.813 | **0.816** |
| | Protective behaviour Before mandate | 0.836 | **0.840** |
| | Protective behaviour After mandate | 0.833 | **0.834** |

| Country | Outcome | RF | XGboost |
|---|---|---|---|
| UK | Face mask wearing Before mandate | 0.755 | **0.760** |
| | Face mask wearing After mandate | 0.695 | **0.705** |
| | Protective behaviour Before mandate | 0.796 | **0.797** |
| | Protective behaviour After mandate | 0.846 | **0.857** |
| India | Face mask wearing Before mandate | 0.757 | **0.764** |
| | Face mask wearing After mandate | **0.856** | 0.851 |
| | Protective behaviour Before mandate | **0.823** | 0.801 |
| | Protective behaviour After mandate | 0.827 | **0.831** |
| US | Face mask wearing Before mandate | 0.846 | **0.853** |
| | Face mask wearing After mandate | 0.855 | **0.862** |
| | Protective behaviour Before mandate | 0.797 | **0.802** |
| | Protective behaviour After mandate | 0.857 | **0.862** |

## 4. Feature importance

Feature importance indicates how much each predictor contributes to the model's prediction. A higher importance score means that the factor plays a stronger role (Calculated after model training based on the contribution of each variable to the predicted results).



# Summary

## Summary of main findings

- **XGBoost** achieved the **best** or joint-best AUC in most modelling tasks, while Random Forest provided a useful comparison model with similar performance in some cases.
- The before/after comparison suggests that **the predictors of compliance are not fixed**. Their importance changes after mask mandates, and the direction of change differs across countries.
- **Some predictors appear repeatedly across countries**, such as non-mask protective behaviour, willingness to self-isolate, survey timing, and confidence-related variables. Other predictors are more country-specific.

## All data and code

All code can be found in my GitHub: https://github.com/ZixaunZhao/COVID-19-Protective-Behavior-Cross-National-Analysis.git